MIL-B-44140(GL)
28 March 1985

MILITARY SPECIFICATION

BEEF STEW, COOKED, DEHYDRATED - (OPERATIONAL RATION COMPONENT)

This specificaiton is approved for use by the Natick Research and Development Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers dehydrated, cooked beef stew for use by the Department of Defense as a component of the Food Packet, Assault.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

EE-B-575 - Bouillon (Soup and Gravy Bases)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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MILITARY

MIL-P-44146 - Packaging of Food Packet Assault Components In Flexible Bags

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Grades of Frozen Carrots

U.S. Standards for Grades of Frozen Peas

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(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

THE UNITED STATES PHARMACOPOEIAL CONVENTION (USP)

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to American Dehydrated Onion and Garlic Association, 251 Kearny Street, No. 410, San Francisco, CA 94108.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

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(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, free from foreign material, evidence of rodent or insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Beef. Beef shall be the only meat ingredient used and shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters and any of the following recognizable primal and sub-primal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds (with heel out). Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins, and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable; and there shall be no evidence of freezing, defrosting, or mishandling.

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3.2.1.1 Boning and trimming. The fresh chilled beef shall be boned and trimmed to remove and exclude objectionable material such as a bone, cartilage, heavy connective tissue, etc. The boneless trimmed meat shall meet the limitations specified in table I and table II.

3.2.1.2 Handling and storage. Handling and storage of boned and trimmed beef prior to processing into the cooked beef logs shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for up to 4 days is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Beef shall not be frozen at any time prior to further processing of the boned and trimmed beef.

3.2.2 Carrots. Carrots shall be fresh or frozen and shall be mechanically diced with dicer settings at approximately 3/8 by 3/8 by 3/8 inch. The carrots shall be Danvers, Red Cored Chatenay or similar varieties with comparable quality suitable for dehydration.

3.2.2.1 Carrots, fresh. Fresh carrots shall be clean, sound, mature and of a good typical color. Fresh carrots shall be washed, peeled, sorted, and trimmed in accordance with good commercial practice.

3.2.2.2 Carrots, frozen. Frozen carrots shall be U.S. Grade A of the U.S. Standards for Grades of Frozen Carrots and shall be of the most recent season's crop.

3.2.3 Onion powder. Onion powder shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products for Standard grade or better onion powder.

3.2.4 Onions, dehydrated, chopped. Dehydrated chopped onions shall comply with the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products for standard or better dehydrated chopped onions.

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3.2.5 Peas, green, frozen. Frozen green peas shall comply with the requirements for U.S. Grade B or better of the U.S. Standards for Grades of Frozen Peas and shall be of the most recent season's crop. The peas shall be of a suitable freezer variety such as Thomas Laxton or Freezer Perfection. The peas shall be split or perforated to improve dehydration and rehydration characteristics.

3.2.6 Potatoes. Potatoes shall be a variety possessing excellent processing qualities and a specific gravity of 0.95 to 1.10. Varieties with these characteristics include Russet, Burbank, Kennebec, Morona, and Norchips. The potatoes shall be washed, sorted, and trimmed to remove defects, and mechanically diced with dice settings at approximately 3/8 by 3/8 by 3/8 inch before being cooked. Dehydrated or dehydrofrozen potatoes shall not be used.

3.2.7 Salt. Salt shall be non-iodized, white, refined sodium chloride, with or without anti-caking agents.

3.2.8 Soup and gravy base, beef flavor. Beef flavored soup and gravy bases shall comply with type I, class I, style A of EE-B-575.

3.2.9 Caramel color. Caramel color shall meet the requirements of the Food Chemicals Codex.

3.2.10 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor and a light color. The white pepper shall contain not less than 1.0 mL volatile oil per 100 grams and be of such a size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.11 Starch, freeze-thaw stable. Freeze-thaw stable starch shall form a smooth, short textured, heavy bodied, almost clear aqueous solution. The flavor of the cooked solution shall be bland with essentially no cereal or starchy taste.

3.2.12 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.13 Nitrogen. Nitrogen shall be U.S. Pharmacopoeia grade and shall be water or liquid nitrogen pumped.

3.3 Preparation and processing.

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3.3.1 Beef preparation.

a. Preparation of beef logs. Beef logs shall be prepared as follows:

1. Coarse grinding and flake-cutting processes. The boned and trimmed meat shall be ground once through a plate having holes 1 to 1-1/2 inches in diameter for coarse grinding or flake-cut by passing it once through a Urschel Comitrol Comminutor equipped with a size 3J030750D head or equivalent. The temperature of the beef shall not exceed 50°F at the end of coarse grinding or 40°F at the end of flake-cutting. Dry ice may be used in appropriate amounts to control temperature during coarse grinding or flake-cutting.

2. Fat content. The fat content of the coarse ground meat shall be 10 to 16 percent. The fat content of the flake-cut meat shall be 12 to 18 percent.

3. Mixing. The coarse ground or flake-cut meat shall be mixed with 0.5 percent salt until the meat becomes moderately tacky then immediately stuffed tightly into suitable casings for cooking.

b. Cooking of beef logs. The cooked beef logs (see 3.3.1.1.b for chilling and freezing conditions) shall be cooked in water or steam until an internal temperature of not less than 150°F and not greater than 160°F is reached.

c. Dicing of cooked beef logs. The cooked beef logs, at a uniform internal temperature of 38°F or lower (see 3.3.1.1 c, d and e for chilling and freezing conditions) shall be mechanically diced with dicer settings at approximately 1/2 by 1/2 by 1/4 inch. After dicing, the beef shall contain a minimum of 75 percent recognizable dices (approximately 1/2 by 1/2 by 1/4 inch).

3.3.1.1 Time-temperature limitations. The beef shall be handled so as to comply with the following limitations:

a. Maximum holding temperature and time of uncooked chilled boneless beef:

	50°F	24 hours

b. Maximum holding temperature of uncooked beef logs:

1. If beef logs are chilled	40°F	24 hours
2. If beef logs are frozen 1/	0°F 2/	30 days

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c. Maximum holding temperature and time of cooked, chilled beef logs for dicing:

38°F 3/	48 hours

d. Maximum holding temperature and time of cooked, frozen beef logs for dicing:

0°F 4/	21 days

e. Maximum holding temperature and time of cooked, tempered (see 6.3) beef logs for dicing:

38°F	48 hours

f. Maximum holding temperature and time of cooked beef dices:

1. If beef dices are chilled	40°F	4 hours
2. If beef dices are frozen 5/	0°F 4/	14 days

1/ If beef logs are frozen in the uncooked state they shall not be frozen after cooking and shall be used in the product within 24 hours after chilling.

2/ The uncooked beef logs shall be frozen to 0°F within 72 hours after placement in the freezer.

3/ The cooked beef logs shall be chilled to 38°F within 8 hours after cooking.

4/ Cooked beef logs and cooked beef dices shall be placed in a freezer within 4 hours after completion of cooking and frozen to an internal temperature of 0°F or below within 48 hours after placing in the freezer.

5/ Beef dices prepared from beef logs that have been frozen prior to cooking or cooked beef logs that have been frozen prior to dicing shall not be frozen.

3.3.2 Preparation of beef stew. The beef stew shall be prepared as preconcentrated and formed slices or as compressed bars. Slice preparation shall be as outlined by 3.3.2.1, 3.3.2.2, and 3.3.2.3. Bar preparation shall be as outlined by 3.3.2.1, 3.3.2.2, and 3.3.2.4.

3.3.2.1 Beef stew formulation. The beef stew shall be formulated as follows:

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Ingredient	Percent by weight
Beef, cooked, diced	25.00
Potatoes, raw, diced	21.00
Peas, green, split or perforated	8.00
Carrots, raw, diced	8.00
Seasoning mix	2.35
Starch, freeze-thaw stable	0.75
Water 1/	34.90

Seasoning mix	Percent by weight
Soup and gravy base, beef flavored	49.4
Onions, chopped, dehydrated	31.9
Salt	12.8
Onion powder	3.4
Pepper, ground, white	1.3
Caramel color	1.2

1/ For preconcentrated and formed slices, the formula water shall be reduced by an amount that will yield a product complying with the finished product requirements.

3.3.2.2 Mixing and cooking. The beef stew shall be mixed and cooked as follows:

a. The raw diced potatoes shall be cooked in boiling water until the potatoes have firm edges and a slightly raw flavor. The potato dices shall be rinsed in cold water to stop cooking and remove starch.

b. The fresh carrots shall be blanched before use in the product.

c. Prepare a slurry with the starch and formula water.

d. Add the seasoning mix to the slurry and mix well. Heat to 180°F with continuous mixing.

e. Add the beef dices, peas, and carrots and heat to 140°F.

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f. Add the precooked and rinsed potatoes.

3.3.2.3 Slice preparation. The preconcentrated and formed slices shall be prepared as follows; or alternatively, the cooked product may be formed so that the dimensions, weights, and volume specified in 3.4(e) are met. The times and temperatures specified below shall be followed according to the process being used.

a. The cooked product shall be immediately mechanically stuffed into suitable casings. The casing shall have an approximate 5-1/2 inch lay flat dimension.

b. Each stuffed casing shall be formed (see 6.4) into a rectangular shape in accordance with 3.4(e).

c. The product shall be placed in a freezer within 4 hours after stuffing and frozen to an internal temperature of 0°F or below within 24 hours after placing in the freezer. Frozen product shall not be held for longer than 30 days before slicing and freeze dehydration.

d. The frozen product shall be cut into slices so that four dehydrated slices will meet the requirements specified in 3.4(e).

e. The slices shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 24 hours after slicing. The product shall be protected at all times from contamination and freezer burn.

3.3.2.4 Bar preparation. The compressed bars shall be prepared as follows:

a. The cooked product shall be portioned onto freezer dryer trays.

b. The product shall be place in a freezer within 4 hours after cooking and frozen to 0°F or lower within 12 hours after placing in the freezer. The product shall be protected at all times from contamination and freezer burn.

c. The product shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 48 hours after placing in the freezer.

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d. The dehydrated product shall be conditioned by adding 10 percent by weight of water. The moistened product shall be processed in a microwave unit at the equivalent of 30 seconds at 1000 watts (see 6.5).

e. The conditioned product shall be compressed into bars 1 inch wide by 3 inches long by not more than 1-1/4 inches thick such that the dehydrated bars will meet the requirements specified in 3.4(f). The compressed bars shall have edges and corners rounded to a nominal 1/8 inch curvature. The product shall be compressed within 5 days of initial freeze dehydration.

f. The compressed bars shall be placed in the freeze dryer and dehydrated as specified in 3.3.3. The dehydration shall be started within 48 hours of compression.

3.3.3 Dehydration. The beef stew shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and product temperatures as indicated by suitable instruments, not to exceed 150°F. If the platen temperature is maintained at 155°F or below with radiant heating, the product temperature may be disregarded. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen. In no case shall more than 16 hours elapse between dehydration and completion of packaging. During this interim period, the product shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches of mercury for the entire period. If vacuum is used, it shall be broken with nitrogen. For compressed bar product during this interim between dehydration and compression the product shall also be protected as above.

3.3.4 Pouch filling and sealing. Each pouch shall be filled and sealed such as to conform to the following requirements:

a. Three bars (compressed) or four slices (preconcentrated and formed) shall be inserted in a mixing bag conforming to MIL-P-44146.

b. The bag closure device (see MIL-P-44146) shall be placed within the outerfolds of the mixing bag so as to minimize any sharp protrusions against the pouch.

c. The filled mixing bag shall be folded and fitted into the pouch.

d. The pouch shall be sealed in accordance with MIL-P-44146.

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3.4 Finished product. The finished product shall comply with the following requirements.

Dehydrated product:

a. There shall be no foreign material such as but not limited to dirt, glass, paint, hard plastic, wood, insects, insect parts or noxious chemicals. There shall be no innocuous soft extraneous material (paper, string, etc.), smears or stains (oil, grease, etc.).

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. There shall be no evidence of incomplete dehydration.

e. When the product is in the form of slices:
   1. Each pouch shall contain four slices.
   2. Each slice shall be approximately 2.25 $\pm$ 0.1 by 3.25 $\pm$ 0.1 by 1/2 to 3/4 inches.
   3. The volume of the four slices shall not be greater than 246 cubic centimeters.
   4. The total weight of the four slices shall be not less than 90 grams and not greater than 100 grams.

f. When the product is in the form of bars:
   1. Each pouch shall contain three bars.
   2. Each bar shall be 3 $\pm$ 0.1 by 1 $\pm$ 0.1 by not greater than 1-1/4 inches.
   3. The total weight of the three bars shall be not less than 90 grams and not greater than 100 grams.
   4. Each bar shall have rounded edges and corners.

g. The product shall be sufficiently compressed or formed so that it can be packaged without breaking or crumbling.

h. Moisture content of the dehydrated product shall not exceed 2.0 percent.

i. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

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j. The E coli count shall be less than 3 per gram (no positives in the the standard 3 tube MPN technique) in four out of five samples and not greater than 20 per gram in any sample.

Rehydrated product:

k. When the product is rehydrated as specified in footnote 3 of table III, the product shall show not less than 70 percent rehydration.

l. The rehydrated product shall possess the flavor, odor, and texture considered normal for frozen and heated beef stew. There shall be no scorched, burnt, or foreign flavors or odors.

m. When rehydrated, the beef stew shall show distinct pieces of beef meat (approximately 1/2 by 1/2 by 1/4 inch), potatoes, carrots and whole peas. The product shall not be a mass of indistinguishable materials.

3.4.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The beef component and the finished product shall originate and be produced; processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with requirements of this document. The cost of grading and acceptance services performed by the USDA involving

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inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot any portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in

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table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories Major AC	Major RE	Minor AC	Minor RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
151		Presence of lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
152		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
153		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
154		Presence of bone measuring 0.3 inch or more in any dimension.
155		Presence of cartilage measuring 0.5 inch or more in any dimension.
156		Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
157		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measure 3 square inches or more.
158		Presence of knuckle cover.
159		Presence of kidney, pizzle eye, prepubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, prepubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in longest dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be determined by a Meat and Poultry Inspection Program employee.

2/ Evidence of freezing or defrosting or presence of any off-condition, (including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration) shall cause rejection of the involved lot.

4.5.1.3 Other ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, handling, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.2.1 In-process testing for fat content. Four 1-pound samples shall be selected at random from the day's production of raw coarse ground beef or

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flaked beef, as applicable. Each of the four samples shall be tested for fat content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. Results shall be reported to the nearest 1 percent. If the fat content of one or more samples is greater than the limit specified in 3.3.1, the lot of ground or flaked beef or any product made therefrom shall be rejected.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44146.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch, mixing bag, and closure device. Any individual net weight of less than 90 grams or greater than 100 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram.

4.5.5 Volume test (slices only). Test for volume of slices shall be performed as follows:

Equipment

a. Container, large enough to submerge pouches in; having a spout approximately 1 inch below the container surface.

b. 150 gm weight with a clip attached.

c. 250 mL or 500 mL beaker.

d. 250 mL or 500 mL graduated cylinder.

d. Vacuum-sealed pouch containing a mixing bag and a closure device.

Procedure

a. Establish tare as follows:

(1) Place container on a level, vibration-free surface.

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(2) Fill the container with water until the water comes out the spout. Wait until the water stops dripping before proceeding.

(3) Position the beaker under the spout.

(4) Attach the vacuum-sealed pouch containing a mixing bag and a closure device to the 150 gm weight with the clip. Submerge the pouch with the attached weight into the filled container.

(5) Allow the water to stop dripping before measuring, approximately 5 minutes.

(6) Measure the collected water in the graduated cylinder and record this measurement as the volume of the tared pouch. Report measurement to the nearest 2 mL.

(7) Empty and dry the beaker and graduated cylinder.

b. Testing of filled and sealed pouch:

(1) Refill the container and, using the same procedure as above, test the filled and sealed pouch.

(2) Measure the collected water in the graduated cylinder and record this measurement as the volume of the filled and sealed pouch. Report measurement to the nearest 2 mL.

c. Determine the volume of the slices using the following equation:

Volume of filled and sealed pouch	minus	Volume of the tared pouch	=	Volume of the slices.

The lot size shall be expressed in filled and sealed pouches. Any result not conforming to the requirement in 3.4(e) shall be scored as a major defect. The sample size shall be the number of pouches indicated by inspection level S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.6 Product examination. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQl, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

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TABLE III. Product defects 1/ 2/

Category		Defect
Major	Minor	
151		Mixing bag missing.
152		Number of bars/slices in pouch not as specified.
	201	Bag closure device missing.
		Applicable to dehydrated product
153		Evidence of incomplete dehydration.
	202	Dimensions of compressed bars not as specified.
	203	Broken or crumbled bar (more than 3 large pieces) slice in a pouch.
	204	Edges and corners of compressed bars not rounded.
		Applicable to rehydrated product 3/
154		Product shows less than 70 percent rehydration. 4/
155		Product does not have distinct pieces of meat (approximately 1/2 by 1/2 by 1/4 inch) potatoes, carrots, and peas (i.e., the product is a mass of indistinguishable materials).

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, noxious chemicals, rancid, sour, stale, etc.), foreign color, or innocuous soft extraneous material, smears or stains (oil, grease, etc.) shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To perform rehydration examination, the entire contents of the pouch shall be used. Break bars or slices each into five or six pieces of comparable size. Add 180° to 200°F water to the broken product in the amount of 4.0 mL of water for each gram of dehydrated product and weigh. Stir to expose all surfaces to the water. Rehydrate for 5 minutes, stir, and rehydrate for an additional 5 minutes. Immediately after the 10 minute rehydration period, pour contents onto a U.S. Standard No. 40 sieve. Drain for 5 minutes and weigh.

4/ Percent rehydration shall be calculated as follows:

$$\frac{\text{Weight of rehydrated product}}{\text{Weight of dry product plus added water}} \times 100 = \text{Percent rehydration}$$

4.5.7 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture-Drying in Vacuum method in the Meat section of the Meat and Poultry Products chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists, except that a temperature of 70°C for 16 hours under a pressure of 100 mm of mercury shall be used. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4(h) shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.8 Microbiological testing. Five filled and sealed pouches of finished product shall be selected at random from the lot regardless of lot size. The contents of each sample pouch shall be tested for aerobic plate count and E. coli in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Microbiological Methods: Section: Examination of Frozen, Chilled, Precooked, or Prepared Foods-Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any result not conforming to the requirements in 3.4(i) and (j) shall be considered a test failure. Any test failure shall be cause for rejection of the lot.

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4.5.9 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44146.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44146.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.4.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Tempering. Tempering shall not be considered freezing and is defined as product chilled below 20°F and tempered up to dicing temperature.

6.4 Forming of slices. Good results have been achieved for forming the preconcentrated slices by placing the casings into screen molds prior to stuffing. Acceptable molds have been found to be those produced by Beacon Metal Products, Inc., of Chicago, IL.

6.5 Compression general. The best results for microwave heating have been achieved by placing the food in a closed container, e.g., a plastic bag, and placing the bagged food flat in the bottom of the oven. If the heating is done in 10 to 15 second cycles, turning the container, and mixing the food between cycles, more thorough distribution of the moisture will be achieved. Products with high fat content may require 1 to 2 minutes to cool before compression.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44140(GL)	Beef Stew, Cooked, Dehydrated - (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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